

ATM Forum signalling protocols and their interworking

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The ATM Forum is a key body which has been developing specifications for asynchronous transfer mode (ATM) technology over the past six years. It has produced a number of signalling specifications covering the user/network interface, the inter-carrier interface and private networking environments. While work continues on 'fine-tuning' these agreements, there is still discussion about how they are used in large-scale networks, and in particular on the interworking of public and private network environments.

This paper reviews the current status of the relevant protocols and their scope (noting that the private network/network interface (PNNI) is covered by Scott and Jones [1]). It then explores the issues related to their interworking and discusses the role of the proposed ATM internetwork interface (AINI) as an alternative internetwork or inter-carrier interface to its broadband intercarrier interface (B-ICI).

1. Introduction

1.1 The ATM Forum's origins, organisation and working method

The ATM Forum was formed in 1991 [2]. It is an international non-profit organisation aimed at accelerating the use of ATM products and services through a rapid convergence of interoperability specifications. It has grown rapidly to its present 900 member companies.

The ATM Forum produces its specifications through debate and discussions within working groups, which report to a technical committee overseeing all technical activity. The ATM Forum technical committee and working groups meet for a week at a time — currently five times per year.

Member organisations may submit 'contributions' for consideration by the working groups at these meetings. Such contributions will typically raise technical issues and propose solutions. If agreed by the meeting, suitable text will be prepared for eventual inclusion in a specification.

The Forum's approach is significant. Being 'contribution-driven' it reflects the imperatives of those members who contribute technical proposals for inclusion during the specification's 'production cycle'. This is in contrast to the approach of the ITU-T, which drives its work with a set of questions at the outset of each study period. There is always the risk with the former approach that the capabilities of a given specification are not as uniformly balanced as might be possible. On the other hand, the process should yield an early specification highly relevant to market needs.

1.2 Terminology

Traditionally, the terms 'private' and 'public' have been used within standards bodies to distinguish a public carrier network — often an incumbent national PTT — from independent, 'in-house' networks. In recent years this clear separation has become blurred by changes in both regulation and the market-place. In some of its discussions the ATM Forum is using the terms 'private network' and 'ATM service provider network' (in place of 'public network') to more accurately reflect the current situation, though the reader will encounter the public/private nomenclature in earlier specifications.

2. The ATM Forum and signalling

Recognising that signalling system requirements differ at different points in the network, the ATM Forum established three working groups to deal separately with each specification together with a fourth to explore interworking issues. This approach, however, was superseded in autumn 1997 by a restructuring along functional rather than interface lines.

2.1 Original structure

The Signalling Working Group

This has dealt with the production of the user/network interface (UNI) signalling specifications. It should be noted here that the UNI specification itself covers many aspects of the interface (such as the physical and adaptation layers) of

which signalling itself forms only a part (albeit a major one). Because the Signalling WG is the main area of signalling expertise, its advice has been sought through joint meetings with other Working Groups where signalling matters arise.

The Broadband Inter-carrier Interface (B-ICI) Working Group

This has addressed the needs of interworking between networks and has been responsible for the B-ICI specification (discussed later). Its scope, however, has concentrated on connection between service provider networks. As will be seen later, the issue of internetwork interfaces, and operation in general, has become a major issue and has not been handled completely within this working group.

The Private Network/Network Interface (PNNI) Working Group

This working group has developed specifications for signalling both within and between private networks. It has been the major focus for routing expertise within the Forum which has been reflected in its approach to the design of its PNNI protocol.

Interworking among ATM Networks (IAN) Working Group

A fourth working group was created during 1997, though strictly it was a joint meeting of the above three working groups. This was the Interworking among ATM Networks Working Group which dealt with a number of issues raised by the growing need to interwork both service provider and private networks in a scalable and coherent manner.

2.2 Revised structure

The ATM Forum restructured these groups in the autumn of 1997. The revised working groups — Control Signalling, and Routing and Addressing — will divide their work along functional rather than interface lines.

3. The signalling specifications in more detail

It has never been the ATM Forum's intention to duplicate work that is already under way; hence, whenever possible, it draws on the output of other standards bodies. However, in practice, it has often been found necessary to add to, or amend the functionality of, existing specifications to support particular requirements.

These considerations are particularly relevant in signalling where much detailed work and output has already been produced. The working groups mentioned above have in each case adopted an ITU-T Recommendation as the

basis of their specifications. However, this has still led to much activity, and additional work where the existing recommendation has not met a perceived need.

In most cases this approach has resulted in specifications that refer back to earlier specifications and recommendations, either noting additional functionality or pointing out differences.

3.1 The Anchorage Accord

A major issue for implementors and users is 'backwards compatibility'. The specification designer is implored to ensure that successive versions of a specification will interwork successfully with earlier versions. This is of course a major confidence factor for implementors and purchasers who would otherwise need to make existing equipment redundant, when upgrading.

To recognise this need, the ATM Forum created the 'Anchorage Accord' at its Anchorage meeting in April 1996. This identified a set of some 60 specifications that it had produced that would form a stable foundation on which to build an ATM network, and encourage backward compatibility [3]. The main signalling specifications included in this set are:

- B-ISDN inter-carrier interface specification v2.0 [4],
- addendum to B-ISDN inter-carrier interface specification v2.0 [5],
- interim inter-switch signalling protocol (IISP) specification v1.0 [6],
- private network/network interface specification v1.0 [7],
- addendum to private network/network interface specification v1.0 for ABR parameter negotiating [8],
- ATM user/network interface (UNI) signalling specification v4.0 [9],
- addendum to UNI signalling 4.0 for ABR parameter negotiation [10],
- user/network interface specification v3.1 [11].

3.2 The user/network interface

The Signalling Working Group have completed a number of specifications relating to the UNI, all based on the concepts embodied in the ITU-T equivalent UNI standard, Q.2931. Of greatest relevance are UNI version 3.1 and UNI version 4.0 (signalling). An earlier ATM Forum standard UNI v3.0 is incompatible with Q.2931 primarily because they use different signalling ATM adaptation layers (SAAL).

Two separate UNIs are identified — a private UNI and a public UNI. The Private UNI exists between a private ATM network and an end-point terminal. The public UNI is used to connect a public ATM network to either an end-point equipment or to a private ATM network. In practice, these two UNIs are very similar.

ATM Forum UNI specification v3.1

UNI v3.1 includes definitions of the physical, ATM and management layers of the UNI. Only section 5 of the specification deals with signalling. UNI v3.1 is broadly a subset of Q.2931, though it includes a number of important enhancements felt necessary for the early deployment and interworking of ATM equipment. The main additions are support for:

- point-to-multipoint connections,
- additional traffic parameters,
- private network addressing.

UNI v3.1 also differs from Q.2931 in the range of message types it supports. Some Q.2931 messages are not supported (ALERTING, PROGRESS, INFORMATION, NOTIFY), while others have been added to support point-to-multipoint working.

ATM Forum UNI signalling specification v4.0

The UNI signalling specification v4.0 was issued in July 1996. Unlike UNI v3.1 it covers purely the signalling aspects of the interface. Again it is based on Q.2931 and is structured to identify differences between itself and Q.2931 rather than being complete in itself.

UNI signalling v4.0 supports the following capabilities which are not covered by UNI signalling v3.1:

- support for the available bit rate (ABR) service,
- parameterised quality of service,
- switched virtual paths,
- narrowband services,
- leaf-initiated join,
- anycast capability,
- frame discard,
- proxy signalling,

- a number of supplementary services (such as both calling and called party number information presentation and restriction, direct dialling in, user-to-user signalling).

Beyond UNI v4.0

The ATM Forum is still active on UNI signalling and is considering further enhancements to ensure compatibility with its other signalling specifications. In January 1997 an Addendum to UNI Signalling V4.0 for ABR parameter negotiation was issued.

3.3 *The private network environment and PNNI*

The ATM Forum has developed PNNI v1.0 (published March 1996) and is working on a further version with additional functionality. PNNI offers both a control signalling protocol and a routing protocol [1]. An early specification for an interim inter-switch signalling protocol (IISP) provided a limited capability but has been superseded by the issue of PNNI v1.0.

The control signalling part of PNNI is based upon the UNI signalling protocol, which is itself based on Q.2931. While this supports point-to-point and point-to-multipoint connections, mechanisms have been added to support source routing, crankback procedures and alternative routing of call set-up requests in the event of set-up failure. PNNI v1.0 does not support some features such as leaf-initiated join.

The routing protocol is based on the link-state routing technique. It is designed to distribute between switches topology information which is then used to compute paths through the network.

3.4 *The inter-carrier interface*

Early work in the ATM Forum was based upon the emerging broadband ISDN user part (B-ISUP), which forms part of the Signalling System No 7 developed by the ITU-T. The model initially assumed by the ATM Forum in its work is one where carrier networks run B-ISUP as an internal network protocol between network nodes.

In the ITU-T view a version of this same protocol then operates between two such networks. From this the ATM Forum has created the broadband inter-carrier interface (B-ICI) protocol, based on the internetwork B-ISUP protocol, to fulfil this role (see Fig 1).

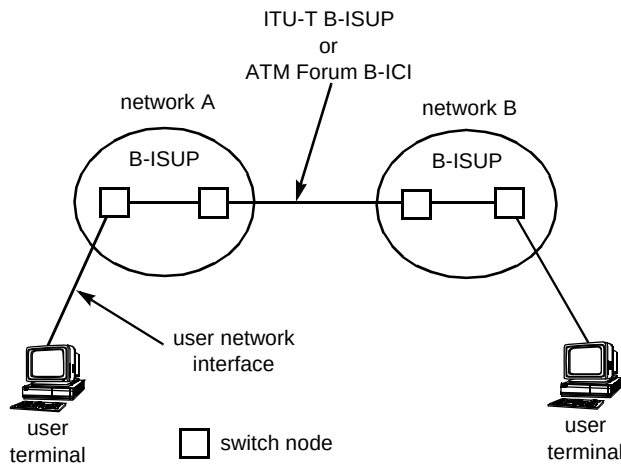


Fig 1 Inter-carrier interfaces to interconnect B-ISUP based networks.

More recently the enthusiastic take-up of the PNNI protocol within the private network environment has led to some proposals within the industry to employ PNNI within carrier or public networks in place of B-ISUP, mainly because of its perceived earlier availability. In this case B-ISUP or B-ICI might not be an appropriate interfacing protocol.

This has led to consideration of an ATM internetwork interface (AINI). This could be used to interconnect two networks where one runs B-ISUP internally, and the other runs PNNI. It might also be used to interconnect two networks both running PNNI, but where full routing interworking is not desired (see Fig 2). This is covered in more detail later.

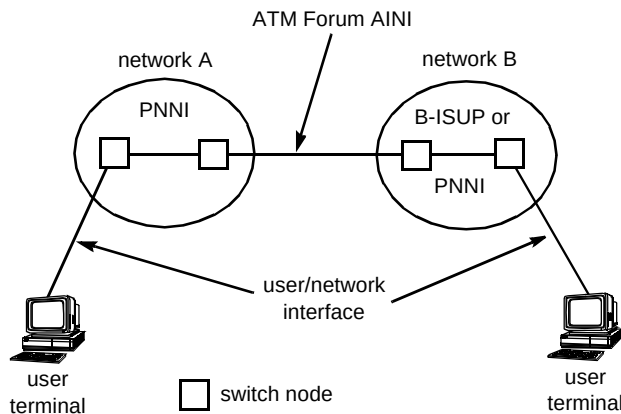


Fig 2 Location of the ATM Forum ATM internetwork interface (AINI).

Two full versions (1.0 and 2.0) of B-ICI have been issued as well as an addendum to v2.0, otherwise known as v2.1. The B-ICI group's method of working has, in general, been to introduce features into the interworking specification, once they have been included in the UNI specification. Hence the latest version of B-ICI has been designed to support the features of UNI v3.1.

Being derived from Signalling System No 7, rather than Q.2931, B-ICI uses a slightly different terminology to both UNI and PNNI signalling. Its signalling messages contain 'parameters' rather than information elements and the call control messages are named differently in many cases.

B-ICI v1.0 was published in August 1993. From a signalling point of view it is of limited significance since it supports only permanent virtual circuits.

B-ICI v2.0, completed in December 1995 introduces switched virtual circuits and much associated functionality. As with UNI v3.1, B-ICI v2.0 contains a full interface definition, hence only section 7 details the signalling capability. Support is offered for both point-to-point connections and unidirectional point-to-multipoint connections.

B-ICI v2.1, an addendum to v2.0 was approved in December 1996 adding support for:

- variable bit rate (VBR) service,
- network call correlation identifier (NCCI),
- ATM end-system addresses (AESAs).

To date, no plans have been announced to produce a further version of B-ICI.

4. Interworking ATM networks

As ATM matures and larger scale networks are envisaged, it will be increasingly important to make sure that the various signalling specifications can be used to allow easy interconnection between networks — both private and ATM service provider. A number of important areas need to be considered.

A range of addressing options exist within the various signalling specifications, some originally intended for the private network domain, some for the public domain. It will be important to ensure that these can be used effectively to achieve call set-ups throughout a concatenation of ATM networks.

Such networks may be interconnected in a number of different ways. These options need to be clearly understood so that routing and control information can be passed satisfactorily as and when required. The choice of interface protocol to connect together private and ATM service provider networks needs also to be considered. A particular case involves the interconnection of two ATM Service

Provider networks when B-ISUP is, for whatever reason, not an option.

4.1 Interconnecting networks

The protocol chosen to interconnect networks must be robust, easily implemented and support as much of the desired functionality as possible. Where both networks operate similar protocols internally, a version of this same protocol may be the obvious choice for interconnection. This is because it will require the minimum of interworking translation between the internal network protocol and the internetwork protocol and support the functionality required as simply as possible. Where, however, the two networks do not run similar internal protocols, some form of interworking will be called for within at least one of the edge nodes. This will require additional functionality with potentially extra cost, and perhaps, development time.

As discussed above, the internal protocols most likely to be encountered within ATM networks at present are PNNI and B-ISUP. Depending on the combination of these encountered, the considerations outlined in the following sections may apply.

4.2 Interconnecting two networks running PNNI

PNNI was developed primarily for use within and between private networks. PNNI's approach to routing involves a logical grouping of the network into peer groups, each with a nominated peer group leader. The process is repeated hierarchically so that a logical tree structure of peer groups may be obtained. A key issue then becomes how exactly the intervening network(s) appears within this topological hierarchy. This will depend in part on whether the two PNNI-based networks wish to exchange topology information.

Interconnecting networks where exchange of topology information is not permitted

If for security or other reasons exchange of topology is not allowed between the networks, the use of the PNNI protocol is not suitable as an interworking protocol. An alternative would be to use the proposed AINI, whose

control signalling functionality would be based on that of PNNI.

Interconnecting networks where exchange of Topology information is permitted

Where exchange of topology information is permitted between the interconnected 'end'-networks, the intervening network(s) may be viewed with varying degrees of complexity by these networks. There are two main ways in which two networks may be interconnected. The simplest is when they are directly connected to each other, without any intervening network. The second case involves interconnection via one or more intervening networks (see Fig 3). The significance of this becomes apparent when one considers the extent to which routing information is passed between the two networks.

At its simplest, the intervening network may appear as a simple link. At its most complex it may be viewed as a network of nodes in its own right, such nodes not necessarily corresponding to the physical ones. These network 'projections' are explored further below.

Four possible 'projections' are identified which can be classified according to how the transit network processes the various items of signalling and routing information. These are summarised as network topology abstractions (see Table 1).

Table 1 Network topology abstractions.

Projection type	Network topology abstraction
Link projection	A single link between the two networks, see Fig 4.
Single node projection	A single PNNI node between the two networks, see Fig 5.
Multiple node projection	Several PNNI nodes between the two networks. The abstraction may not map one-to-one on to the physical nodes interconnecting the two networks (see Fig 6).
Node for node projection	This is a literal one-to-one projection and may be of limited applicability.

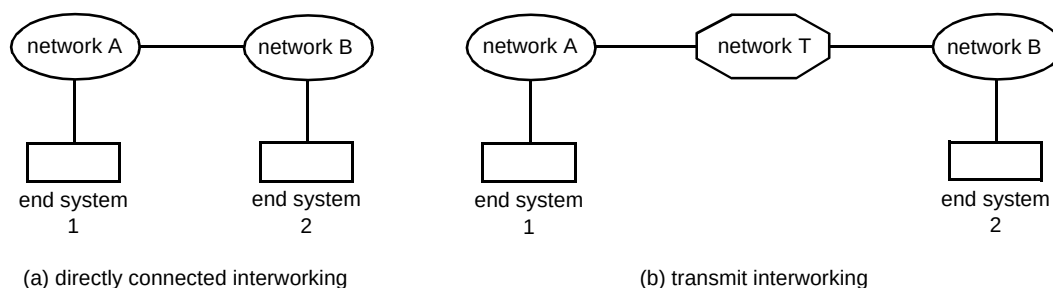


Fig 3 Interworking configurations.

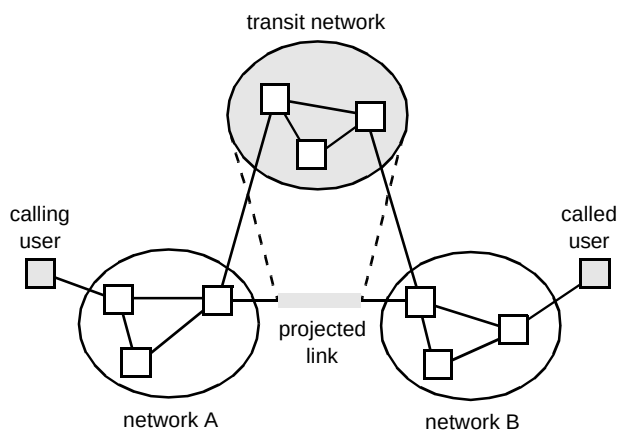


Fig 4 Link projection.

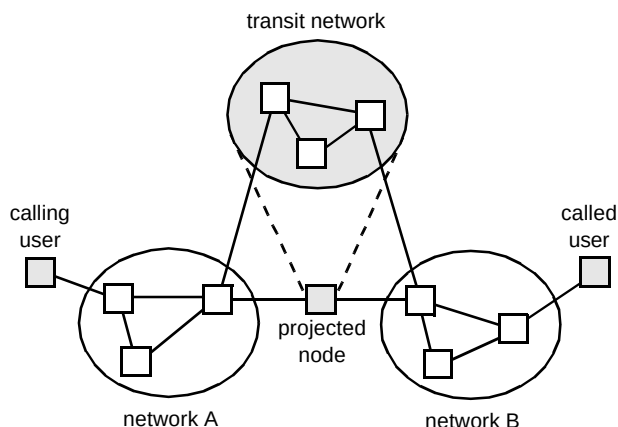


Fig 5 Single node projection.

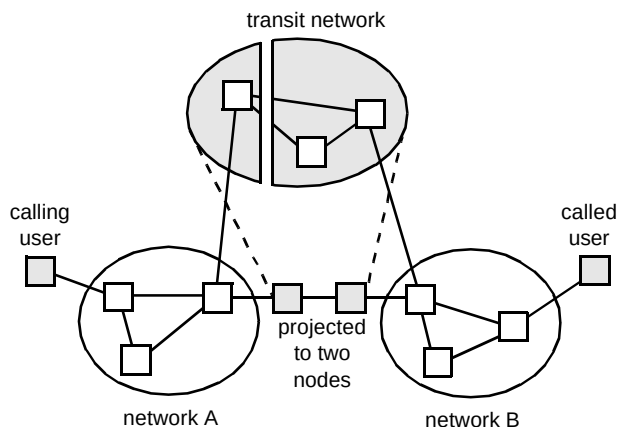


Fig 6 Multiple node projection.

Functionality of the intervening network

In addition to the way in which the intervening network is projected, the extent of support it offers must also be considered. At least four scenarios can be identified (see Table 2) which differ according to whether and how the intervening network handles the following information:

- basic signalling, i.e. the basic call control without the PNNI-specific items,
- PNNI signalling — the specific signalling items associated with PNNI (such as DTLs),
- PNNI routing protocol.

Scenario 1 is effectively a tunnelling solution where no signalling support is offered by the intervening network.

Table 2 PNNI peering scenarios.

Scenario	Protocol support with intervening network		
	Basic signalling	PNNI signalling	PNNI routing
1	✗	✗	✗
2	✓	✗	✗
3	✓	✗	✓
4	✓	✓	✓

4.3 Interconnecting networks — both B-ISUP based

Where both networks to be interconnected use B-ISUP as an internal protocol, then the ATM Forum's B-ICI protocol is appropriate for this purpose. Being itself based on B-ISUP this approach will support maximum functionality across the network boundary.

4.4 Interconnecting networks — a mix of PNNI and B-ISUP based

This is the situation which suggested the need for the development of the ATM internetwork interface. The interface design would need to support as much functionality as possible consistent with compatibility of the two protocols. PNNI is being widely implemented in switches, because of their deployment in private networks. Hence it has been suggested that, to avoid the burden of such switches having to implement a second (B-ISUP) protocol stack, the AINI be based on the existing PNNI protocol, but without the routing protocol aspects of PNNI since these would not be required.

5. Addressing

Addressing has become of particular importance as the interworking of service provider and private networks has developed. Historically the various B-ICI, PNNI and UNI signalling protocols each developed their own statements on allowable addressing structures and methods. These reflected the needs perceived at the time by the specification developers, but did not offer guidance on how they should be used in an interworking scenario.

Current ATM Forum protocols support a number of address formats. The main division is between 'native' E.164 numbers and ATM end-system addresses (AESAs). The former have evolved out of the telephony environment via the ITU-T while AESAs have been primarily data-oriented.

5.1 Native E.164 addresses

A native E.164 address may be up to 15 digits in length and comprises:

- a country code,
- a national destination code
- a subscriber number.

While Q.2931 and B-ISUP recommendations allow other 'type of number' formats, the ATM Forum protocols allow only the international format.

5.2 AESAs

AESAs are derived from the ISO network service access point addresses. There are currently three types of AESA supported by the ATM Forum in existing specifications:

- data country code (DCC),
- international code designator (ICD),
- E.164-type,

each being 20 octets in length. The various types each comprise an initial domain part (IDP) and a domain specific part (DSP). The IDP defines which type of address it is by specifying an administration authority responsible for assigning values to the remaining part of the address (the DSP).

A particularly important feature of addressing for scalability to large networks is that addresses be aggregatable in a sensible hierarchical manner. If not, a flat, non-hierarchical scheme will lead to large routing tables within ATM switches.

5.3 Consequential impact of addressing philosophy

Some features and supplementary services which utilise addresses will require careful attention to ensure that interworking is facilitated where different address types are employed. An example is the closed user group (CUG) capability which defines a group of users and the permitted connectivity both within and outside the group. This uses addressing information as part of the identification mechanism. A further example — the network call correlation identifier (NCCI) — is specified in B-ICI

version 2.1. These are used to uniquely identify a specific call over a period of time, for purposes such as charging.

The ATM Forum recognises the importance of ensuring that practical interworking solutions for addressing are well understood and is currently devoting considerable time to the issue.

6. Other issues

A number of key issues — for example addressing and the InterCarrier interface have already been covered. There are, as well, some other areas relevant to the development of ATM signalling standards.

6.1 Support for supplementary services

Support for supplementary services — such as are defined for voice services and narrowband ISDN — will be key for the full benefit offered by broadband networks to be realised. Where these are carried across networks using a range of protocols it will be important to ensure that the protocol interworking provides a consistent service as viewed by the user regardless of the network implementation.

6.2 Multipoint-to-point connections

Point-to-point connections are naturally the simplest to implement. Point-to-multipoint have followed, while aspects of multipoint-to-point are now under discussion. As well as aspects of merging streams of cells, there are issues associated with routing and combining different qualities of service which need to be addressed

6.3 Mobile and wireless ATM

Signalling and routing protocols will play an important part in the development of effective mobile and wireless applications of ATM. In particular there has been discussion about the extent to which routing philosophies developed within the PNNI specification work can be exploited in the design of call routing and hand-off procedures in the mobile environment.

6.4 Interaction with internetworking protocol (IP)

IP routers may be interconnected over an ATM network by means of a simple network of permanent virtual circuits (PVCs). This potentially makes inefficient use of the ATM network bandwidth, and requires establishment by the network management capability. A particular area of interest involves exploring more efficient and interactive methods of supporting IP traffic within an ATM environment. Two such methods have been suggested — PNNI augmented routing (PAR) and integrated PNNI (I-PNNI).

PAR involves routers with ATM interfaces advertising their status into the PNNI network, receiving similar information from other such routers, and using the information to establish SVCs across the PNNI network. These could then be used to interconnect routers more efficiently.

While PAR maintains a separation between the PNNI and IP routing protocols, I-PNNI seeks to use a single routing protocol — PNNI — both for routers and ATM nodes. The ATM network routing information is effectively extended beyond the ATM network itself.

7. Conclusions

The ATM Forum membership has put much effort into developing specifications to accelerate the roll-out process of ATM switch technology. While much of its output has been based on existing ITU-T Recommendations, it has extended these in its PNNI specification by the inclusion of routing. A major challenge for the future will be to ensure that the signalling systems can interwork appropriately to meet operator requirements for manageable and scalable networks.

Many of the key items mentioned above are still under discussion at the time of publication. Given the energy with which the ATM Forum pursues such issues, they are likely to evolve considerably in the coming months. The reader should therefore regard the foregoing as a snapshot of the current situation.

References

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